ECE 329 **Homework 4** **Due: Sat, Oct 18, 2025, 11:59PM**

Totaling 80pts with 20pts extra Bonus Points. (Bonus problems are optional and the earned bonus credits will be used to "fill in" the "missing points" in the total homework score.)

1. (10pts) **Vacuum diode problem:** Consider a simplified model of a **vacuum diode** consisting of a **cathode** in the $x = 0$ plane and an **anode** in the $x = d$ plane, where the anode is held to a constant potential $V_a = 3$ V relative to the cathode. The region $0 < x < d$ between the cathode and the anode supports a charge density $\rho(x)$ accounting for the electrons in transit from the cathode (where they are emitted) to the anode (where they are captured). If the potential distribution in the region $0 < x < d$ is given by $V(x) = V_a(x/d)^{4/3}$, find the following:

   a) (3pts) Electric field $\mathbf{E}$ at $x = d/2$,

   b) (3pts) Volumetric charge density ρ at $x = d/4$,

   c) (4pts) The surface charge density ρ_s on the anode.

2. (10pts) **Potential distribution between two plates:** Consider two infinite, plane parallel, perfectly conducting plates at $z = 0$ and $z = z_o > 0$, which are kept at potentials $V = 0$ and $V = V_p > 0$, respectively. The region between the plates is filled with two slabs of perfect dielectric materials having permittivities ϵ_1 for $0 < z < d$ (region 1) and ϵ_2 for $d < z < z_o$ (region 2).

   a) (5pts) Find the solutions for the electric potentials, $V = V(z)$, in the two regions by solving Laplace's equation and enforcing the continuity of $V(z)$ at $z = d$. **Hint:** $V(z) = Az$ for $z < d$ and $V(z) = Ad + B(z - d)$ for $z > d$ where parameters A and B depend on V_p as well as $\epsilon_1$and ϵ_2.

   b) (5pts) Given that $z_o = 3d = 3$ m, $V_p = 6$ V, $\epsilon_1 = \epsilon_o$, and $\epsilon_2 = 3\epsilon_o$, what is the surface charge density ρ_s of the conductor plate at $z = z_o$?

3. (10pts) The gap between a pair of parallel infinite copper plates extends from $z = 0$ to $z = W > 0$ and is originally occupied by vacuum (ϵ_o, μ_o). The plates carry equal and oppositely signed surface charge densities and as a consequence we have a constant electric field $\mathbf{E} = 4\hat{z}$ V/m in vacuum in the gap region.

   a) (2pts) What are the corresponding displacement vector $\mathbf{D}$ and polarization vector $\mathbf{P}$ in the gap region?

   b) (2pts) What is the surface charge density ρ_s of the copper plate at $z = 0$?

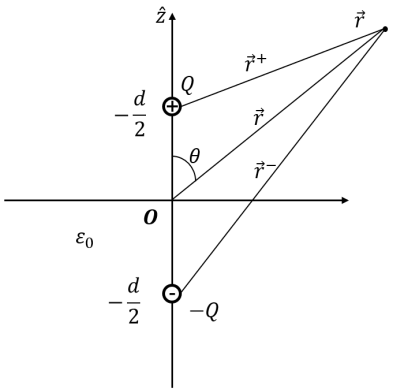
   c) (3pts) What would be the new steady-state values of $\mathbf{E}$, $\mathbf{D}$, and $\mathbf{P}$ in the gap region if the gap were filled with a non-conducting fluid of permittivity $\epsilon = 81\epsilon_o$ without changing the surface charge densities of the copper plates?

   d) (3pts) What would be the new equilibrium values of $\mathbf{E}$, $\mathbf{D}$, and $\mathbf{P}$ in the gap region if some amount of salt were dissolved in the fluid in the gap (see part c) to raise its conductivity to $\sigma = 4$ S/m (conductivity of sea water)? State the values of $\mathbf{E}$, $\mathbf{D}$, and $\mathbf{P}$ after a steady-state equilibrium is reached and briefly explain your answer.

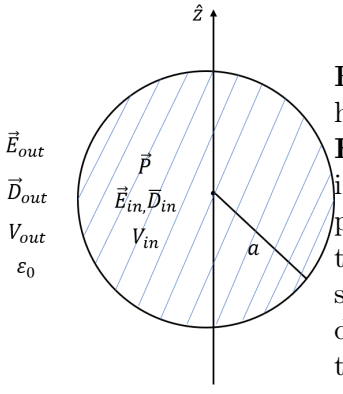
4. (10pts) Consider the following spherically symmetric configuration of composite materials in steady-state equilibrium:
(i) The region defined by $r \leq a$, where $r = \sqrt{x^2 + y^2 + z^2}$ is the radial distance from the origin, has $\epsilon = \epsilon_o$, $\sigma = 10^6$ S/m, and holds a net charge of $Q = +4$ C.
(ii) A perfect dielectric shell with $\epsilon = 4\epsilon_o$ occupies the region $a < r < b$.
(iii) Region $b \leq r \leq c$ has the same material properties as region $r \leq a$ and holds a net charge of -1 C.
(iv) Region $r > c$ is occupied by free space.

In all four regions we have $\mu = \mu_o$. Determine, in all four regions, (a) $\mathbf{D}$, (b) $\mathbf{E}$, and (c) $\mathbf{P}$, *and* (d) the surface charge densities, in C/m^2 units, at each of the three material boundaries at $r = a$, b, and c.

Hint: Make use of Gauss's law in integral form, $\oint_S \mathbf{D} \cdot dS = \int_V \rho dV$, with $\mathbf{D} = \epsilon\mathbf{E} = \epsilon_o\mathbf{E} + \mathbf{P}$, and a crucial fact about steady-state fields within conducting materials.



(a)



(b)

Bonus (20pts)**:** In class we have discussed the induced field $\mathbf{E_p} = -\mathbf{P}/\varepsilon_0$ due to the polarization $\mathbf{P}$ inside a dielectric slab perpendicular to $\mathbf{P}$. We argued that this is not a universal result. In this example, we try to derive the induced field $\mathbf{E}_p$ due to the polarization $\mathbf{P}$ confined in a spherical region.

5.

a) (10pts) First let us consider the electrostatic potential V due to a dipole moment $\mathbf{p} = \hat{\mathbf{z}}Qd$ centered at the origin in free space with permittivity ε_0, as illusrated in Fig. (a).

i. (5pts) Please represent $V(\mathbf{r})$ in terms of Q, $\mathbf{r}^+ = \mathbf{r} - \frac{d}{2}\hat{z}$ and $\mathbf{r}^- = \mathbf{r} + \frac{d}{2}\hat{z}$. Assume the potential at infinity $V(\infty) = 0$.

ii. (5pts) Please make an approximation to the result $V(\mathbf{r})$ of (a) when $r \gg d$. **Hint:** Note $r^+ \to r - \frac{d}{2}\cos\theta$, $r^- \to r + \frac{d}{2}\cos\theta$.

b) (10pts) Consider the case when a uniform polarization $\mathbf{P} = P\hat{z}$ is confined in a spherical region $r < a$ centered at the coordinate origin in free space with permittivity ε_0, as illustrated in Fig. (b). The polarization $\mathbf{P}$ generates $\mathbf{E}_{in}$ and $\mathbf{E_{out}}$ inside and outside the spherical region, respectively. $\mathbf{E}_{in}$ is associated with displacement field $\mathbf{D_{in}}$ and electrostatic potential V_{in}, and $\mathbf{E}_{out}$ is associated with displacement field $\mathbf{D_{out}}$ and electrostatic potential V_{out}.

i. (2pts) Assume $V_{in}(\mathbf{r}) = Az$, where A is a coefficient to be determined. Note this is a solution to $\nabla^2 V_{in} = 0$. Please represent $\mathbf{E}_{in}$ and $\mathbf{D_{in}}$ using A and P.

ii. (3pts) In view of the results in part (a), we can assume $V_{out}(\mathbf{r}) = B\frac{\cos\theta}{r^2}$, where B is another coefficient to be determined. Note this is a solution to $\nabla^2 V_{out} = 0$. Please represent $\mathbf{E}_{out}$ and $\mathbf{D_{out}}$ using B. **Hint:** the gradient operator in spherical coordinate is $\nabla V = \mathbf{r}\frac{\partial V}{\partial r} + \hat{\theta}\frac{1}{r}\frac{\partial V}{\partial \theta} + \hat{\phi}\frac{1}{r\sin\theta}\frac{\partial V}{\partial \phi}$.

iii. (2pts) Please make use of V being continuous at $r = a$ and the boundary condition of $\mathbf{D}$ at $r = a$ to solve the unknown coefficient A in terms of P, and represent $\mathbf{E}_{in}$ in terms of $\mathbf{P}$.

iv. (3pts) Please show that the boundary condition of $\mathbf{E}$ at $r = a$ and the fact that V being continuous at $r = a$ lead to the same constraints on A and B. **Hint:** The constant unit direction vector $\hat{z}$ can be represented in terms of the position dependent unit direction vectors $\hat{r}$ and $\hat{\theta}$ as $\hat{z} = \hat{r}\cos\theta - \hat{\theta}\sin\theta$.

6. (10pts) **Continuity eqn and transient response in conductors:**

a) (2pts) Show that in a homogeneous conductor where $\mathbf{J} = \sigma\mathbf{E}$, Gauss's law $\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_o}$ and the continuity equation $\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} = 0$ can be used together to derive a differential equation

$$\frac{\partial \rho}{\partial t} + \frac{\sigma}{\epsilon_o}\rho = 0$$

for the charge density ρ. **Hint:** this was already done in class last week, also it is in the class notes.

b) (2pts) Find the solution to the differential equation above for $t > 0$ if at $t = 0$ the charge density is $\rho(x, y, z, 0) = \cos(5x)$ C/m^3 over all space.

c) (2pts) According to the solution found in part (b), how long would it take for ρ to reduce to $0.01\cos(5x)$ C/m^3? Assume that $\sigma = 5.8 \times 10^7$ S/m.

d) (2pts) Given the solution to part (b) for $\rho(x, y, z, t)$ and Gauss's Law $\nabla \cdot \mathbf{E} = \rho/\epsilon_o$ find an electric field solution $\mathbf{E}(x, y, z, t)$ having only an x-dependent x-component. **Hint:** First determine $\partial E_x/\partial x$ and then find its anti-derivative.

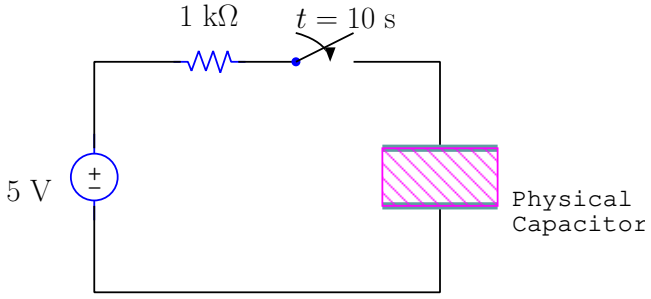
e) (2pts) Based on part (d) what is $\mathbf{J}(x, y, z, t)$ as $t \to \infty$?

7. (10pts) **Capacitance, conductance - practical case:** A parallel plate capacitor has a $d = 5$ mm plate separation, square shaped plates with 2 cm X 2 cm dimensions and a dielectric filling with a permittivity of $\epsilon = \pi\epsilon_o$ and conductivity of $\sigma = 10^{-3}$ S/m.

   a) (2.5pts) Determine the capacitance $C = \epsilon A/d$ of the physical capacitor and express it in pF units. **Hint:** pF=10^{-12} F

   b) (2.5pts) Determine the conductance $G = \sigma A/d$ of the same capacitor and express the corresponding resistance $R = 1/G$ in kΩ units. **Hint:** kΩ =$10^3\,\Omega$.

   The capacitor described above is now placed in a circuit shown below where a switch action at $t = 10$ s connects the capacitor across 5 V source with 1 kΩ internal resistance:



   c) (2.5pts) If the capacitor described is holding $Q = 1$ C charge at time $t = 0$ (prior to closing the switch in the above circuit), what will the capacitor charge be at $t = 30$ ns? **Hints:** (i) ns=10^{-9} s, (ii) redraw an equivalent circuit model where the physical capacitor is represented as a parallel combination of an ideal capacitor C from part (a) and an ideal resistor R from part (b), (iii) remember that $Q = CV$ and $dV/dt + V/(RC) = 0$ in the circuit with the switch open and the capacitor is discharging through R in series with C!

   d) (2.5pts) Determine the capacitor voltage at $t = 20$ s, which is 10 s later than the switch action. **Hint:** how does 10 s compare to the time constant $R_{th}C$ in the circuit where $R_{th}$is the Thevenin resistance of the network seen by the capacitor.

8. (10pts) **Spherical capacitor:** Consider a pair of metallic spherical shells of radii a and $b > a$ with their centers coinciding with the origin of the reference coordinate system. The medium between the shells has permittivity $\epsilon = 2\epsilon_o$ and conductivity $\sigma = 2 \times 10^{-6}$ S/m.

   a) (2 pts) Use the integral form of Gauss's law to relate the static field between the shells to charge Q residing on the (outer) surface of the inner shell with radius a and integrating the field from radius a to b obtain the relation $Q = CV$ where V is the potential drop from the inner to outer shell and
   $$C = 4\pi\epsilon \frac{ab}{b-a}.$$

   b) (2 pts) Taking the limit of capacitance C as $b \to \infty$, obtain the capacitance (in pF) of a metallic shell of radius $a = 1$ m embedded in an infinite region of permittivity $\epsilon = 2\epsilon_o$ and conductivity $\sigma = 2 \times 10^{-6}$ S/m.

   c) (2 pts) What is the conductance (in μS) of the metallic shell configuration described in (b). **Hint:** make use of $G = \frac{\sigma}{\epsilon} C$ relation from class notes.

   d) (2 pts) Develop an RC equivalent circuit to represent the physical system described above. Use this equivalent circuit to derive a differential equation that governs the temporal variation of the charge $Q = CV$ on the spherical shell.

   e) (2 pts) Assume $Q(t) = 1$ C on the shell with radius a at $t = 0$. The radial electric field produced by $Q(t)$ will drive a radial current to discharge the shell. Obtain an expression for $Q(t)$ in $t > 0$ that describes the discarge of the metallic shell by solving the differential equation obtained in (d).

9. (10pts) **Copper wire:** Copper is a highly conducting metal with a **conductivity** of $\sigma = 5.8 \times 10^7$ S/m and a free-electron density of $N_e = 8.45 \times 10^{28}$ m^{-3}.

   a) (2.5pts) Determine the resistance R of a copper wire of radius $r = 1$ mm and length d =200 m using
   $$R = \frac{1}{G} = \frac{d}{A\sigma}$$
   from Lecture 10. A denotes the cross-sectional area of the wire.

   b) (2.5pts) What would be the magnitude of electric field **E** within the wire of part (a) if the wire were conducting a 1 A current? You may assume a uniform current distribution across the wire cross section.

   c) (2.5pts) With the electric field determined from part (b), what would be the mean speed $|\mathbf{v}|$ of an electron in the wire? **Hint:** first deduce the current density $\mathbf{J} = \sigma\mathbf{E}$ in the wire, and then use the fact that $\mathbf{J} = N_e q\mathbf{v}$ in the copper wire, with $q = -e = -1.6 \times 10^{-19}$ C denoting the charge of an electron.

   d) (2.5pts) How long would it take the electron in part (c) to drift from one end of the wire to the other? **Hint:** you may be surprised by the result of your calculation.